



Certyfikat PMP: Project KPI Analysis

dr Zbigniew Galar

The Core Mechanism

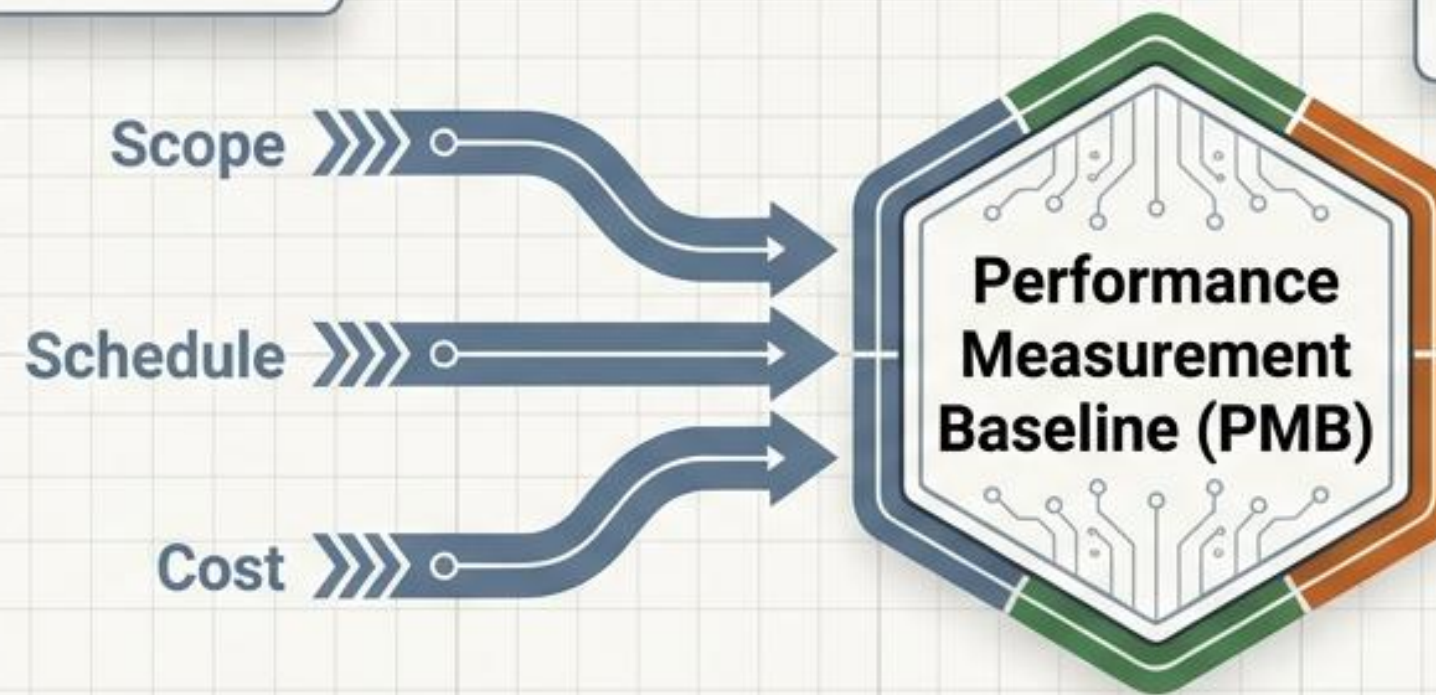
Earned Value Management (EVM) continuously monitors and compares what has been received or produced against what has been spent.



The PMB Integration

Combines **Schedule**, **Cost**, and **Scope** baselines into a single **Performance Measurement Baseline (PMB)**.

Note: Includes contingencies, but excludes management reserves.



The Diagnostic Goal

Determine variance causes, assess impact on the project, and determine the exact corrective actions required.



⚠ Cheat Code

PV = What we planned.

EV = What we did.

AC = What we paid.

Planned Value (PV)

The approved budget assigned to work to be completed during a given period.

Alt Acronym: BCWS
(Budgeted Cost of Work Scheduled)

Earned Value (EV)

The value of the work completed to date compared to the budget.
(Cannot exceed the PV budget for the activity).

Alt Acronym: BCWP
(Budgeted Cost of Work Performed)

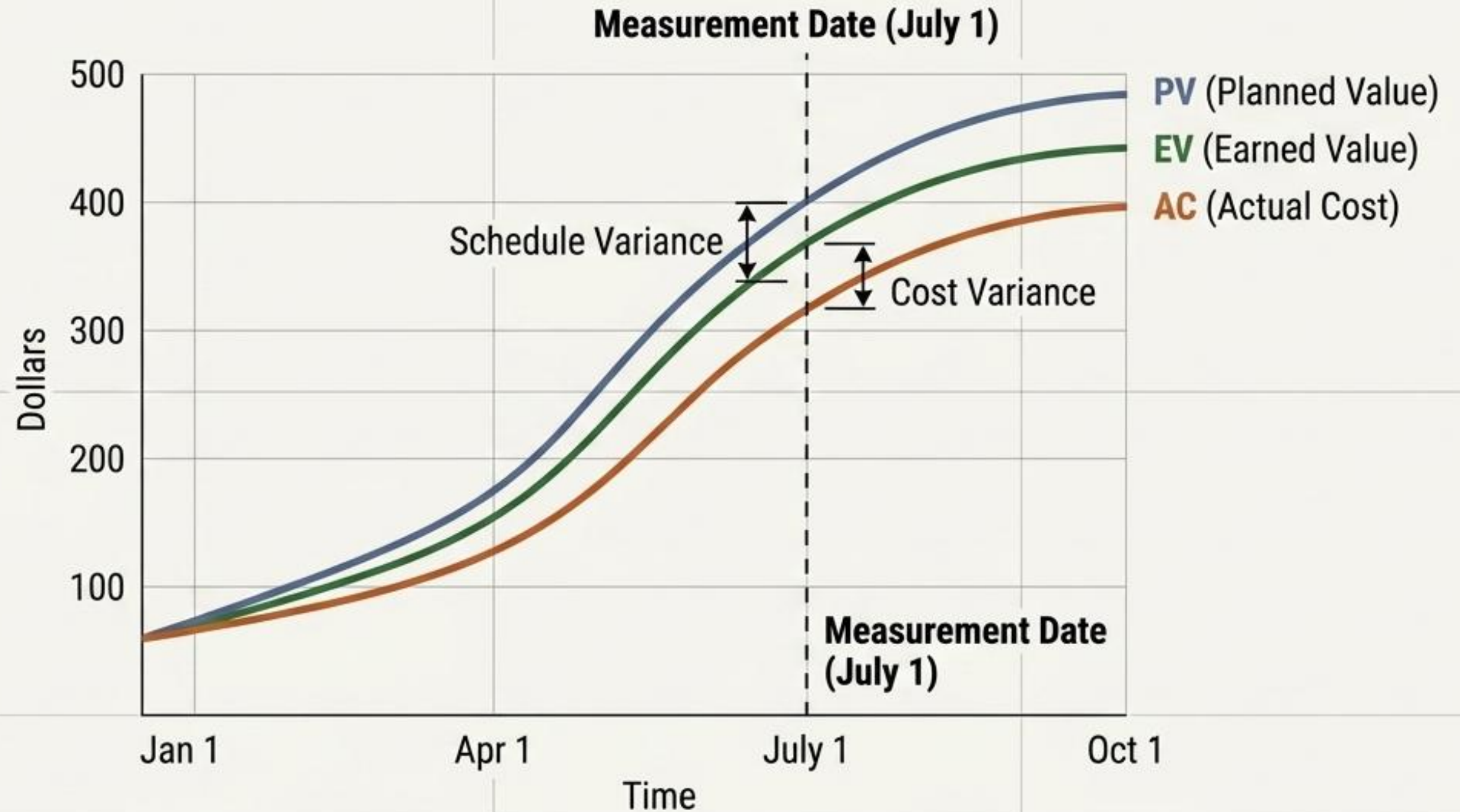
Actual Cost (AC)

Money actually expended during a given time period for completed work.

Alt Acronym: ACWP
(Actual Cost of Work Performed)

Earned Value Management: S-Curve Analysis

Dashboard Readout (July 1)
PV = 400
EV = 375
AC = 325



Mathematical formulas are simply representations of physical gaps on a timeline.



CHEAT CODE:

Negative Variance = BAD (Overrun/Behind)

Positive Variance = GOOD (Under budget/Ahead)

Cost Variance (CV)

Formula: $CV = EV - AC$

$$375 - 325 = +50$$

Positive (+). Spending less than planned for the work completed.

Schedule Variance (SV)

Formula: $SV = EV - PV$

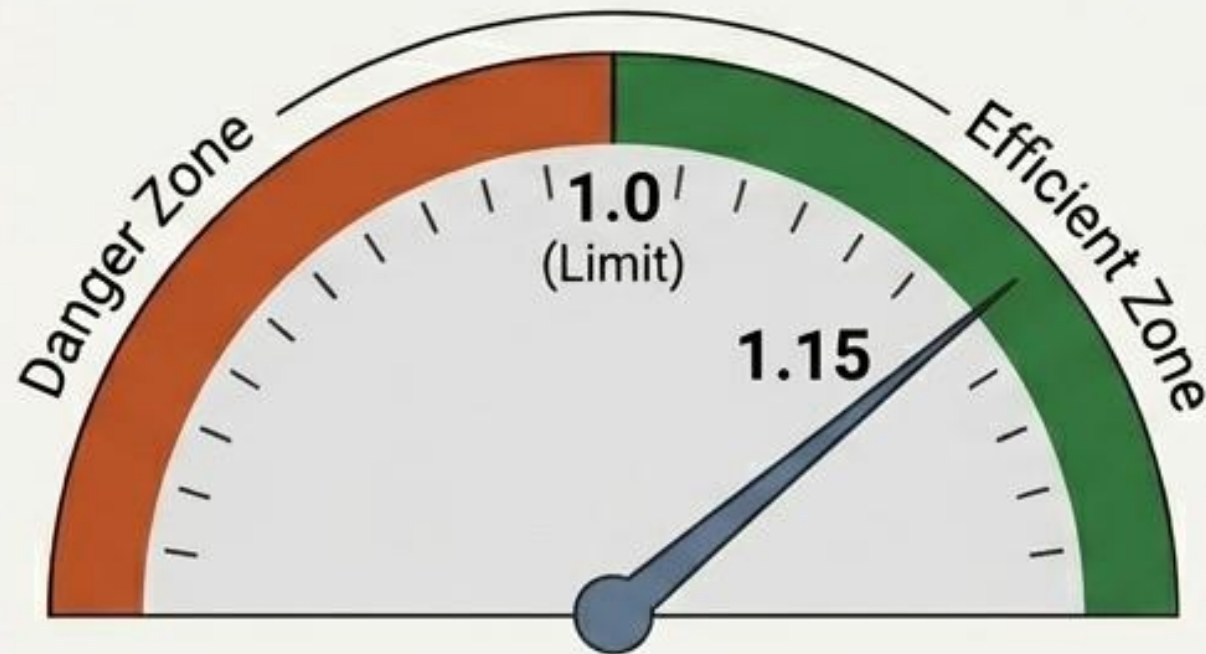
$$375 - 400 = -25$$

Negative (-). Behind where we planned to be.

Phase 2 Diagnostics: Relative Efficiency

! CHEAT CODE:
Index > 1.0 = GOOD
Index < 1.0 = BAD

Cost Performance Index (CPI)

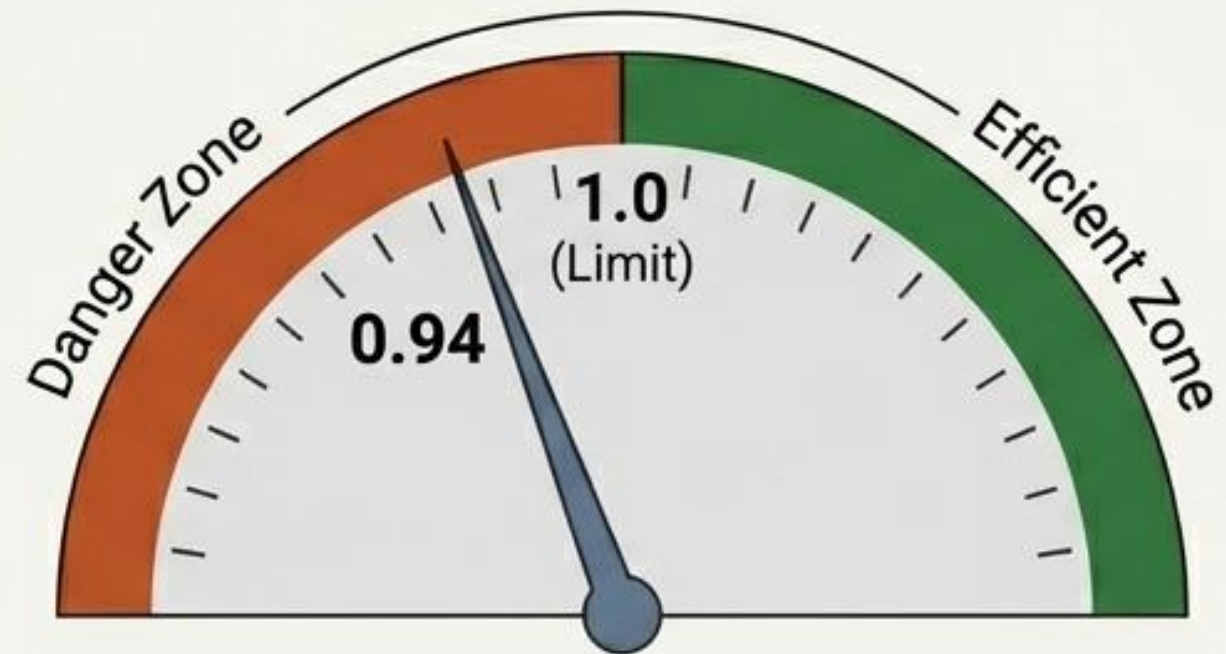


Formula: $CPI = EV / AC$

$$375 / 325 = 1.15$$

A+ Performance

Schedule Performance Index (SPI)



Formula: $SPI = EV / PV$

$$375 / 400 = 0.94$$

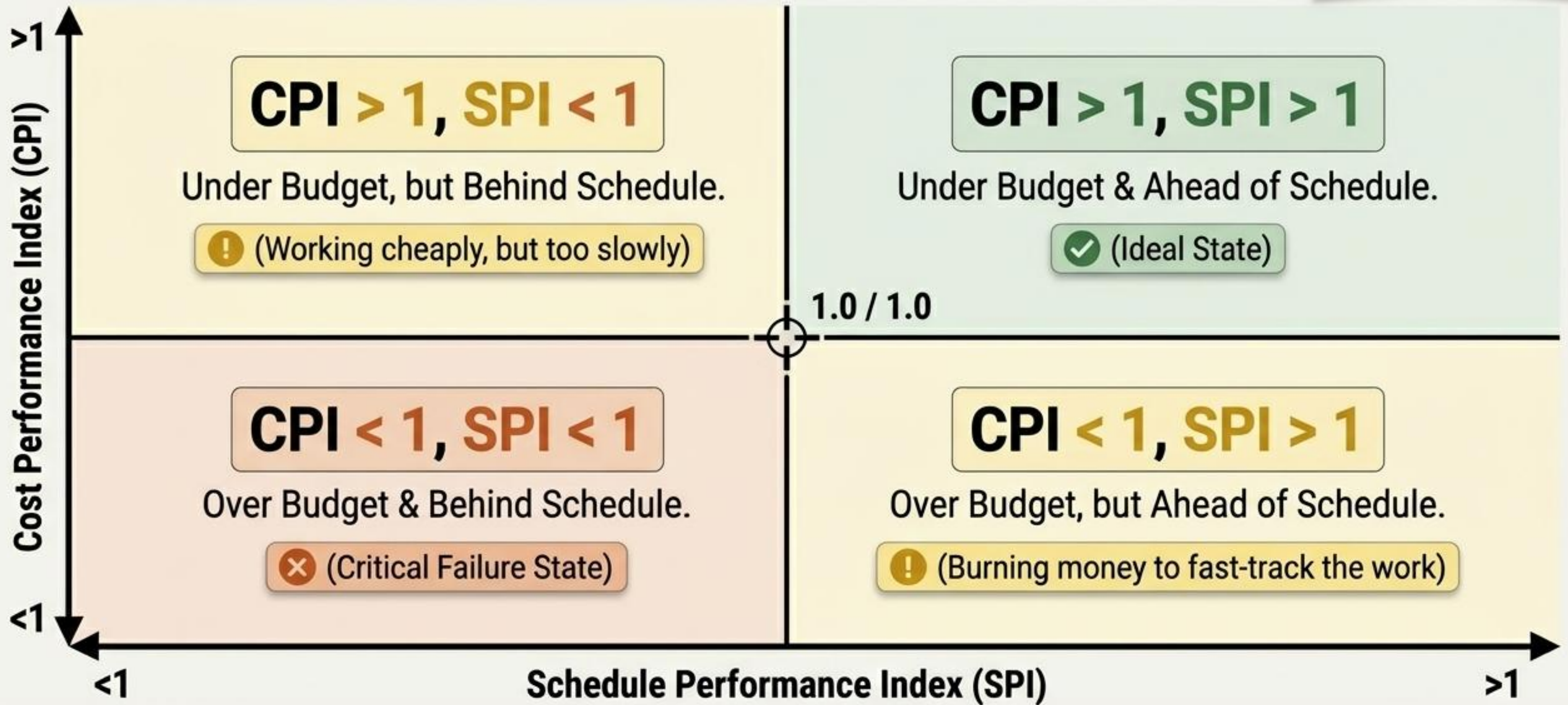
Losing Efficiency

	Absolute Variance	Relative Index
Cost	$CV = EV - AC$ Answers: How much money are we off by?	$CPI = EV / AC$ Answers: How efficiently are we spending our budget? Most Critical PMBOK Metric
Schedule	$SV = EV - PV$ Answers: How much schedule volume are we off by?	$SPI = EV / PV$ Answers: How efficiently is the team completing work tasks?

Cumulative CPI/SPI Note: Calculated using the sum of all EV, AC, and PV to predict performance at project completion.

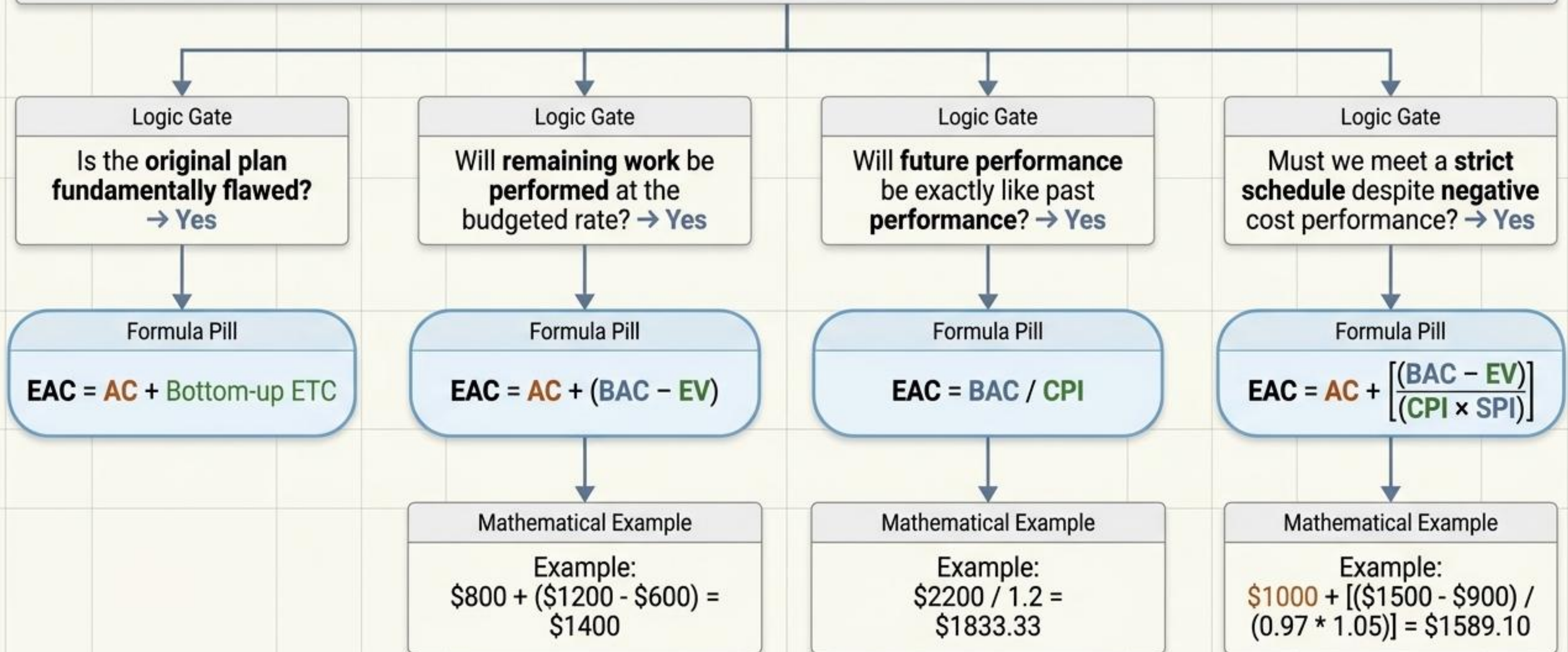
Phase 3 Diagnostics: Comprehensive Performance Grid

! CHEAT CODE:
Index > 1.0 = GOOD
Index < 1.0 = BAD



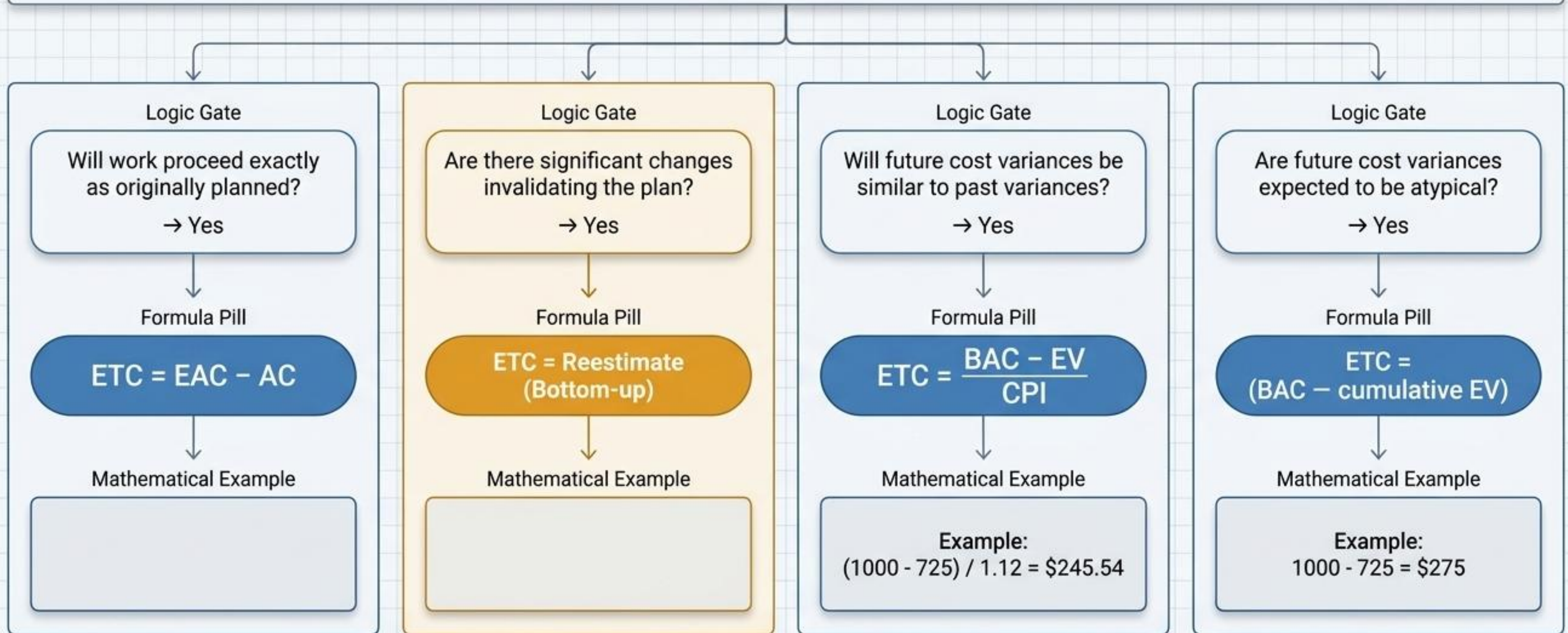
Cumulative CPI/SPI Note: Calculated using the sum of all EV, AC, and PV to predict performance at project completion.

Core Question: What will the total project cost be? (EAC)



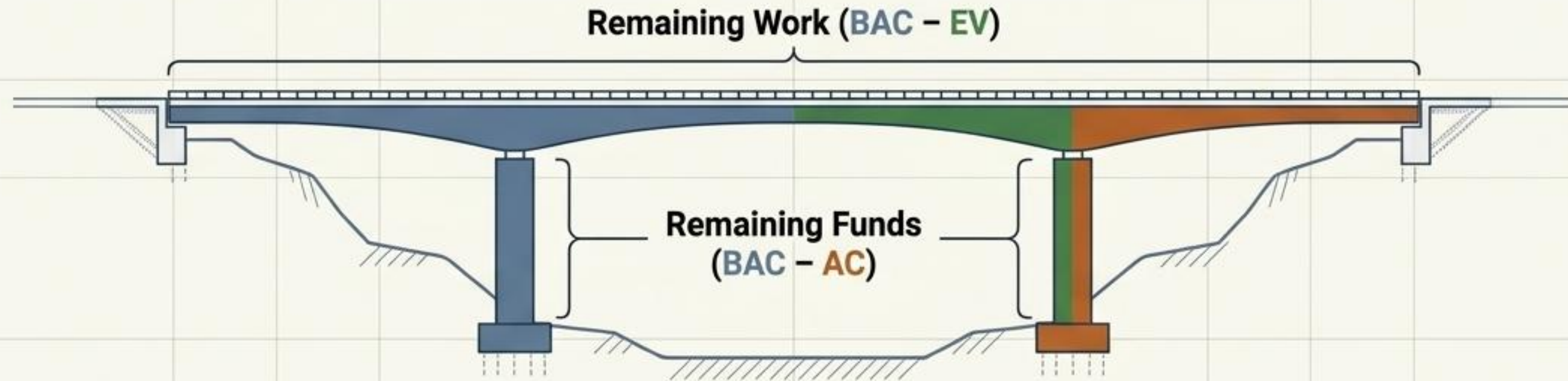
Cumulative CPI/SPI Note: Calculated using the sum of all EV, AC, and PV to predict performance at project completion.

Core Question: How much more funding is needed to finish? (ETC)



Cumulative CPI/SPI Note: Calculated using the sum of all EV, AC, and PV to predict performance at project completion.

To-Complete Performance Index (TCPI): The projected cost performance the remaining work must achieve to hit the goal.



Scenario A (Hitting original BAC)

$$\text{TCPI} = \frac{\text{BAC} - \text{EV}}{\text{BAC} - \text{AC}} \rightarrow$$

Example: Need 1.5x efficiency to hit goal.
Requires drastic improvement.

Scenario B (Hitting revised EAC)

$$\text{TCPI} = \frac{\text{BAC} - \text{EV}}{\text{EAC} - \text{AC}} \nearrow$$

Example:
New goal set, requires 0.75 efficiency.

⚠ The Golden Rule: If cumulative CPI falls below 1, all future work must perform at the TCPI rate to stay within the authorized budget.

Target Landing & Variance Analysis (VAC)



Variance Analysis (VAC)

Calculates the final projected difference between the original budget and the new estimate.

Formula: $VAC = BAC - EAC$

[-] Negative Number = Final cost overrun.

[+] Positive Number = Final cost underrun.

Trend Analysis Note

Forecasts project outcomes based on historical information. As project performance improves over time, variances should systematically become smaller, moving landing points closer to the central target.

MASTER SYNTHESIS: The Diagnostic Framework

	Metric Category	Specific Variables	The Strategic Question Answered
1	The Vitals	PV, EV, AC	What is the raw baseline data? ●
2	Variances (Absolute)	CV, SV	How much are we off by? ●
3	Indexes (Relative)	CPI, SPI	How efficiently are we working? ●
4	EAC (Prognosis)	4 Formulas	What will the total project cost?
5	ETC (Funding)	4 Formulas	How much more money do we need?
6	TCPI (Correction)	BAC / EAC Goals	How hard must we work to hit the target?
7	VAC (Final Outcome)	BAC - EAC	Where will we ultimately land?